RAMDEOBABA

Idea lab

COURSE CODE

JANNIVARAN

AI-Powered Citizen Grievance Redressal & Tracking System

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PROBLEM STATEMENT

The Problem

- Citizens often face difficulty in registering and tracking public complaints.
- Complaints may experience delays, poor visibility, or lack of follow-up.
- Citizens may not know:
- Who is handling the complaint
- What its current statusis

Who is Affected?

- Citizen s
- Administrative authorities

Why is it Important?

- Atransparent grievanceworkflow can improve complaint visibility, response monitoring, and accountability.
- The JANIVARAN specifically focuses on preventing lost complaints and delayed resolutions through routing, SLA tracking, notifications, and escalation

EXISTING SOLUTION / LITERATURE REVIEW

Existing Solutions

- Traditionalcomplaintregistration through:
- Helpl ines
- Email
- Department-specific portals

Limitations

- Limited visibility into the complete complaint lifecycle
- Manual follow-up may still be required
- Citizen assistance during complaint submission may be limited

Research / Market Gap

- Needfora unified platformthat combines:
- Complaint Filing→ Department Handling → SLA Monitoring → Escalation → Notifications → Resolution Feedback

Proposed Approach

- JANIVARAN combinesthesefunctions into one citizen + administration workflow, including visual status tracking, SLA monitoring, role-based access, automated notifications, and AI support

PROPOSED SOLUTION

- JANIVARAN — Citizen Grievance Redressal Portal

1. Citizen Portal

- Secure login/authentication
- Online complaint submission
- Photo evidence upload
- AI support chat

2. Adm inistrative Portal

- Centralized complaint dashboard
- Monitor total, pending and breached complaints
- Complaint resolution anddeletioncontrols

3. Aut omated Workflow

- Submit → Review → In Progress → Resolve

4. Smart Monitoring

- SLAbreach identification
- Automa ted escalation
- Email notifications with timestamps

OBJECTIVES

01 — Digitalize Grievance Registration

- Provideasimpleonlineplatformforcitizenstosubmitcomplaints with supporting evidence.

02 — Enable Transparent Tracking

- Allowcitizenstomonitorcomplaintprogressthroughavisual timeline and SLA deadline.

03 — Improve Administrative Accountability

- Provideofficialswithcentralizeddashboards,role-basedaccess, status management, and escalation mechanisms.

04 — Automate Communication

- Sendautomated notificationswhenevercomplaintstatuschanges.
- Unique complaint identification
- Real-time status visibility

JANIVARAN

People's Voice, Stronger Tomorrow

SYSTEM ARCHITECTURE / METHODOLOGY

Architecture of JANIVARAN – Citizen Grievance Redressal System

Report • Track • Resolve
for a Better Tomorrow

Citizen (User)

- Register / Login
- Submit Complaint (with details & images)
- Track Complaint Status
- View Timeline & SLA
- Receive Notifications
- Provide Feedback

Access Web Portal

View Updates

Frontend (Web Application)

Next.js 14 + React + Tailwind CSS



Manage Complaints

View Reports

Admin / Authority

- Login to Admin Dashboard
- View & Manage Complaints
- Assign to Departments
- Update Status / Resolve
- Monitor SLA & Reports
- Escalate Overdue Complaints

Backend (Next.js API Routes)

Business Logic | Authentication | Complaint Processing | Notifications

Firebase Authentication
User Authentication
(Role-based Access)

Cloud Firestore (Database)
Store Complaints
User Data | Status | Logs

Notification Service
Email Notifications
(SMTP / Nodemailer)

Technology Stack

- **Next.js 14** (Frontend & API Routes)
- **React** (UI Library)
- **TypeScript** (Programming Language)
- **Tailwind CSS** (UI Styling)
- **Firebase Authentication** (User Management)
- **Cloud Firestore** (Database)
- **Nodemailer / SMTP** (Email Notifications)
- **React Context API** (State Management)

Workflow / Methodology

1. **User Registration** — Citizen registers/ logs in using Firebase Authentication
2. **Submit Complaint** — Fill details, category, location and upload evidence
3. **Store in Database** — Complaint stored in Firestore with unique Tracking ID
4. **Processing & Assignment** — Auto classification, SLA generation and assigned to department
5. **Monitoring** — Track status, SLA and escalate if overdue
6. **Resolution** — Status updated, notification sent to citizen
7. **Feedback** — Citizen provides rating and feedback

Goal | A secure, scalable and transparent platform to ensure timely resolution of citizen grievances through technology.

IMPLEMENTATION

- Software Technologies
- Next.js 14
- React
- Tailwind CSS
- Firebase Authentication
- Google Firestore
- Firebase Admin SDK

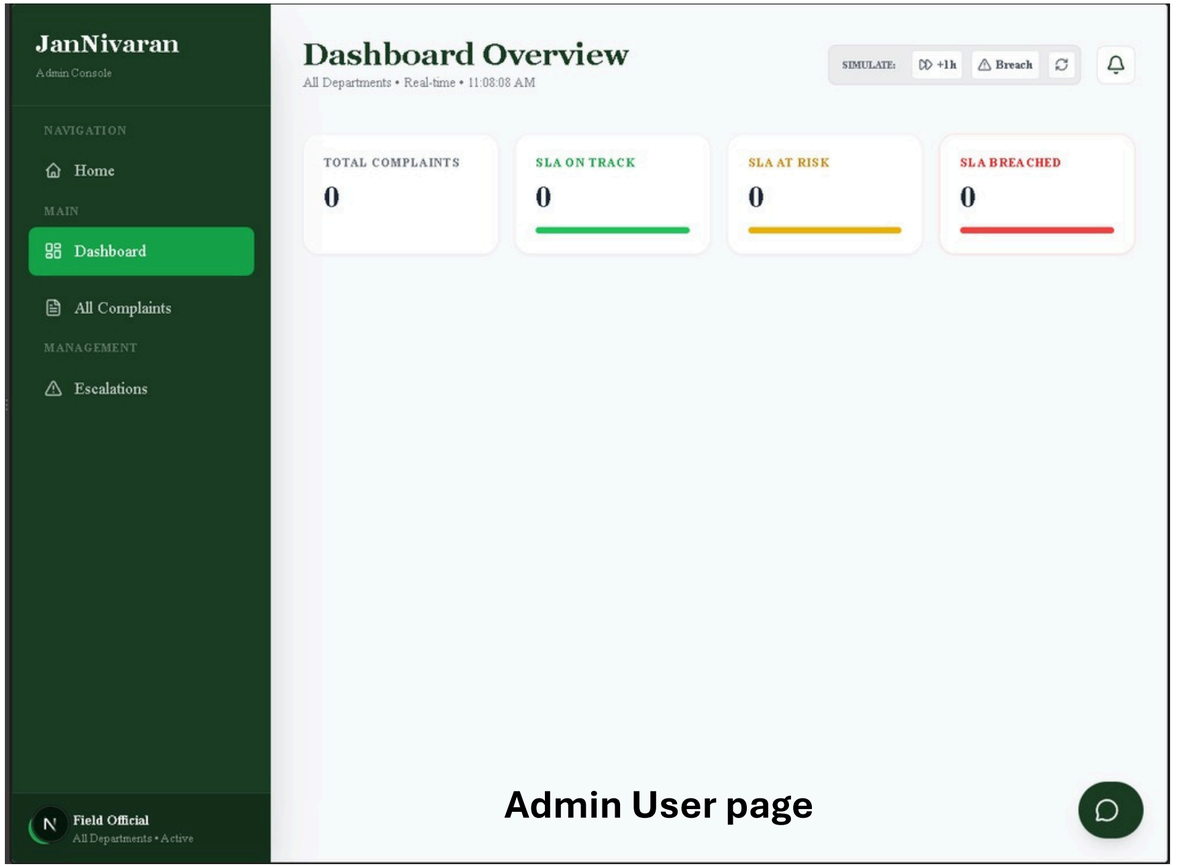
MajorImplementedModules

- Citizen registration/login
- Complaint submission
- Photo evidence upload
- Complaint tracking dashboard
- Status timeline

RESULTS / PROTOTYPE



Client User Base With AI Chatbot



Admin User page

INNOVATION, IMPACT, DELIVERABLES & FUTURE SCOPE

Innovation

- Combinescomplaint management + SLA monitoring + escalation + notifications
- Visual complaint lifecycle
- AI-based citizen support

Impact

- Grea ter complaint visibility
- Better monitoring of delayed cases
- Reduced dependence on manual follow-up

Deliverables

- Com plaint tracking system
- Notification mechanism
- AI support module

Future Scope

- Mobile application
- Multilingual / regional-language support
- Predictive prioritization of complaints
- SMS / WhatsApp notifications

CONCLUSION & REFERENCES

Conclusion

JANIVARANprovides a centralized digital platform for citizen grievance registration, tracking, monitoring, and resolution.

Key Takeaways

- Simplifies complaint registration
- Provides transparent real-time status tracking
- Supports role-based administrative processing
- Enables automated notifications and escalation

Reference :

Government grievance-redressal systems

1. CPGRAMS — Centralized Public Grievance Redress and Monitoring System